

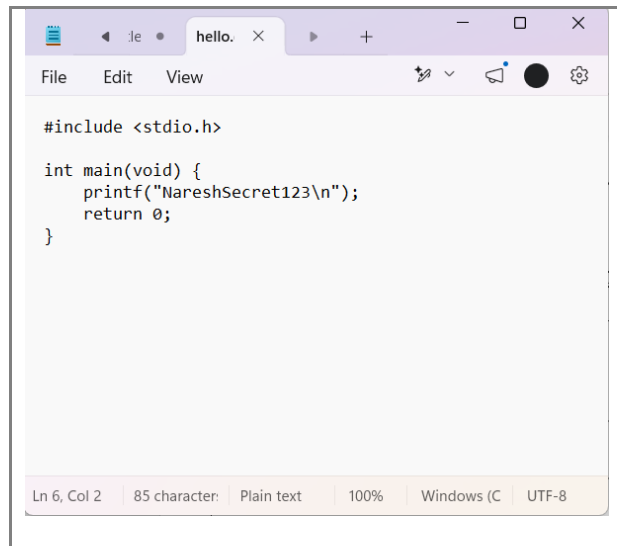
**AIM:** To perform basic reverse engineering on a small C program by compiling it to a Windows executable and analysing it using the 'strings' utility and the Ghidra decompiler to recover its logic.

**PROCEDURE:**

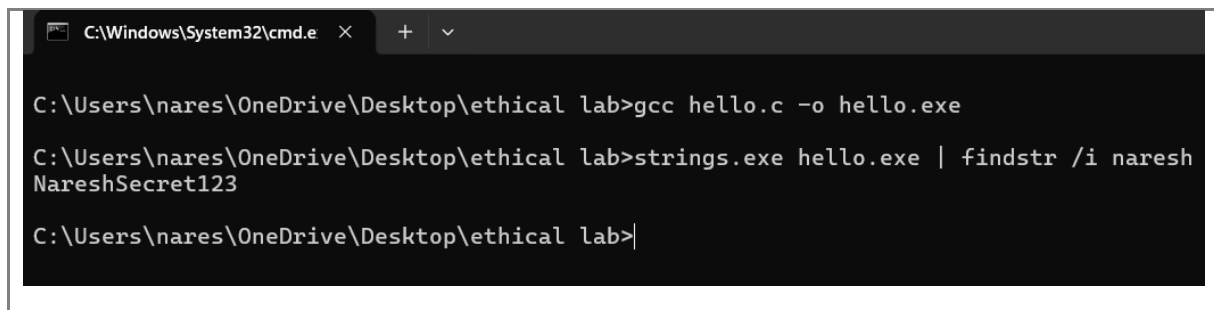
1. Create a small C source file hello.c with the following content using Notepad:

```
#include <stdio.h>

int main(void) {
    char secret[] = "NareshSecret123";
    printf("Hello Ethical Hacker!\n");
    return 0;
}
```

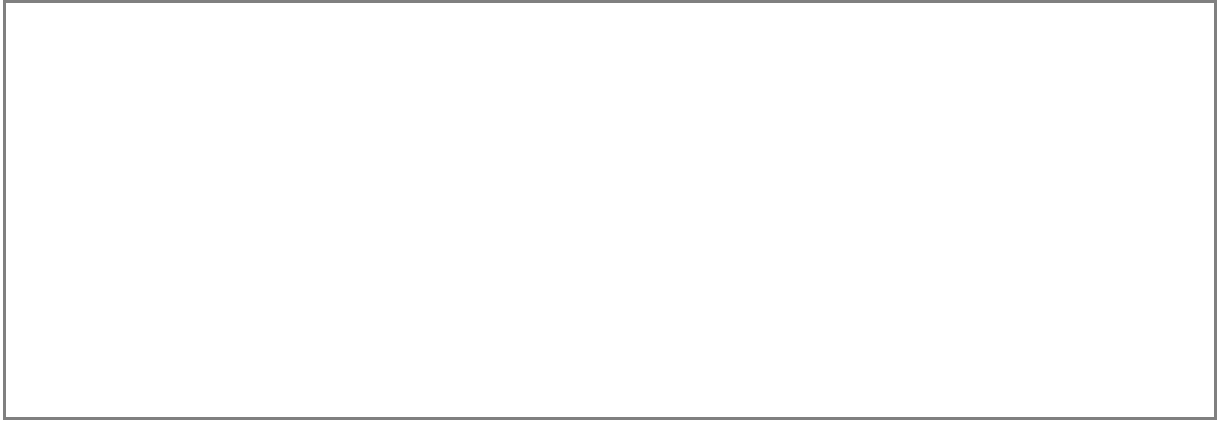


2. Compile the source file using MinGW-w64 with the command 'gcc hello.c -o hello.exe' inside Command Prompt.



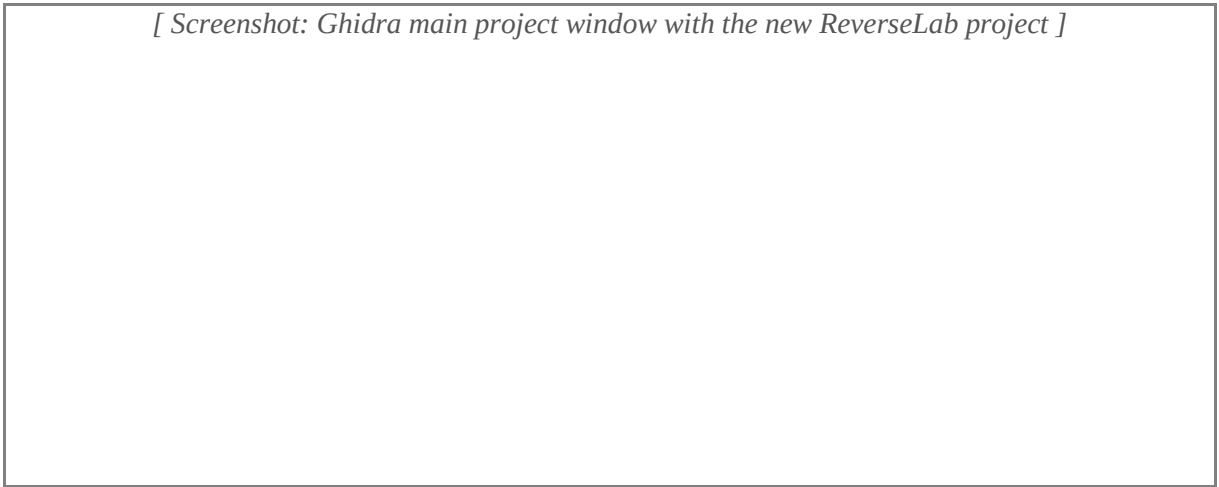
3. Download the standalone strings.exe from Sysinternals (<https://learn.microsoft.com/sysinternals/downloads/strings>) and run 'strings.exe hello.exe | findstr /i naresh'. Observe that the embedded string 'NareshSecret123' is leaked from the compiled binary.

*[ Screenshot: Strings tool output showing the embedded NareshSecret123 string ]*



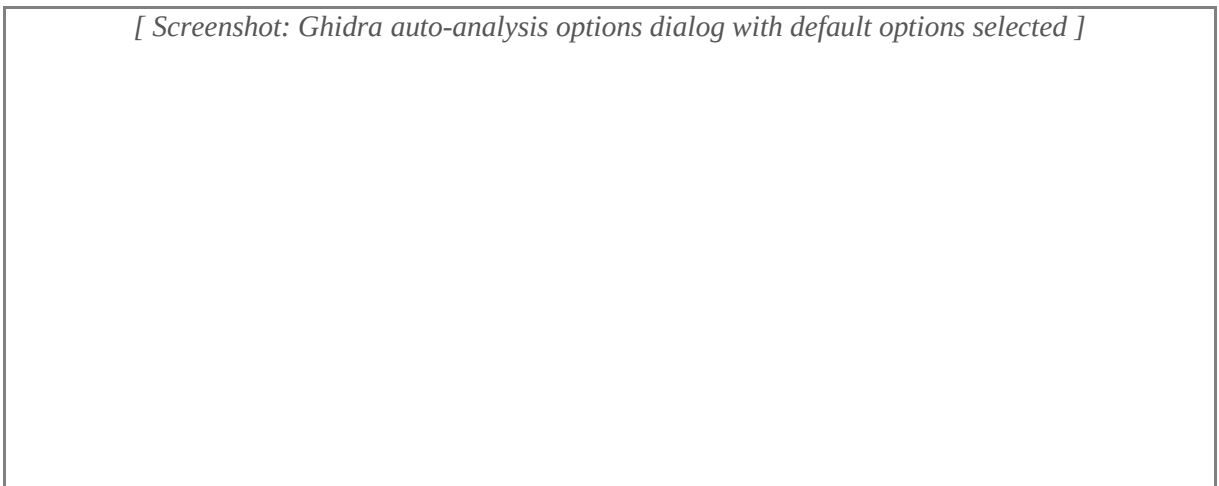
4. Download Ghidra from <https://ghidra-sre.org> and extract it. Launch ghidraRun.bat to open Ghidra and create a new project named ReverseLab.

*[ Screenshot: Ghidra main project window with the new ReverseLab project ]*



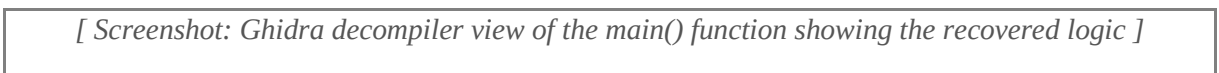
5. In Ghidra, drag and drop hello.exe into the project, double-click it and accept the default analysis options. Wait for the auto-analysis to complete.

*[ Screenshot: Ghidra auto-analysis options dialog with default options selected ]*



6. In the Symbol Tree, open the 'main' function. The decompiler window on the right displays a high-level C-like reconstruction of the program, recovering the original printf call and the embedded secret string.

*[ Screenshot: Ghidra decompiler view of the main() function showing the recovered logic ]*





**RESULT:** Thus, reverse engineering of a small Windows executable was performed successfully using the strings utility and the Ghidra decompiler, and the embedded data and program logic were recovered.